

Python Variables & Data Types — Complete Class Notes

1. What is a Variable?

A variable is used to store data in memory so that we can use it later in the program.

Syntax

```
variable_name = value
```

Example

```
name = "Gnanesh"  
age = 24  
salary = 35000
```

2. Rules for Variable Names

Correct Variable Names

```
student_name = "Ravi"  
age2 = 20  
mySalary = 50000
```

Wrong Variable Names

```
2name = "Ravi"  
student-name = "Ravi"  
my salary = 50000
```

Variable Naming Rules

- Variable names should not start with numbers
- Spaces are not allowed
- Special symbols are not allowed except _
- Variable names are case-sensitive

3. Integer (int)

Integer stores whole numbers without decimal values.

Example

age = 24
marks = 100

Features

- No decimal point
- Positive and negative numbers supported

4. Float

Float stores decimal values.

Example

price = 99.99
temperature = 36.5

Features

- Contains decimal values
- Used for prices, percentages, measurements

Difference Between int and float

int	float
Whole numbers	Decimal numbers
10	10.5

5. String (str)

String stores text data.

Example

```
name = "Python"  
city = 'Hyderabad'
```

Features

- Written inside quotes
- Can contain letters, numbers, symbols

String Operations

Concatenation

```
first = "Hello"  
second = "World"  
  
print(first + second)
```

Repeat String

```
print("Hi " * 3)
```

6. Boolean (bool)

Boolean stores only:

- True
- False

Example

```
is_logged_in = True  
is_payment_done = False
```

Uses

- Login systems
- Payment status
- Conditions

7. List

List stores multiple values in ordered form.

Example

```
fruits = ["apple", "mango", "banana"]
```

Features of List

- Ordered
- Changeable
- Duplicates allowed

List Methods

append()

Adds value at end.

```
fruits.append("grapes")
```

insert()

Adds value at specific position.

```
fruits.insert(1, "orange")
```

remove()

Removes specific value.

```
fruits.remove("apple")
```

pop()

Removes last value.

```
fruits.pop()
```

len()

Returns total values count.

```
print(len(fruits))
```

8. Tuple

Tuple stores ordered but unchangeable values.

Example

```
colors = ("red", "green", "blue")
```

Features of Tuple

- Ordered
- Not changeable
- Duplicates allowed

Difference Between List and Tuple

List	Tuple
Changeable	Not changeable
[]	()

List	Tuple
------	-------

append() possible append() not possible

9. Set

Set stores unique unordered values.

Example

```
numbers = {1,2,3,3,4}
```

Output

```
{1,2,3,4}
```

Features of Set

- Unordered
- No duplicates
- Changeable

Set Methods

add()

```
numbers.add(5)
```

remove()

```
numbers.remove(2)
```

Difference Between List and Set

List	Set
Ordered	Unordered
Duplicates allowed	Duplicates not allowed

List	Set
Indexing possible	Indexing not possible

10. Dictionary (dict)

Dictionary stores data in key-value pairs.

Example

```
student = {  
    "name": "Ravi",  
    "age": 20,  
    "marks": 95  
}
```

Features of Dictionary

- Key-value pair format
- Ordered
- Changeable

Access Dictionary Values

```
print(student["name"])
```

Add New Value

```
student["city"] = "Hyderabad"
```

Update Value

```
student["age"] = 21
```

Remove Value

```
student.pop("marks")
```

Difference Between Set and Dictionary

Set	Dictionary
Only values	Key-value pairs
No keys	Keys available

11. type() Function

Used to identify datatype.

Example

```
print(type(10))
print(type(10.5))
print(type("Python"))
```

12. Type Conversion

int()

Converts value into integer.

```
x = int("10")
```

float()

Converts value into float.

```
y = float(5)
```

str()

Converts value into string.

```
z = str(100)
```


13. Complete Data Type Comparison Table

Data Type	Ordered	Changeable	Duplicates
List	Yes	Yes	Yes
Tuple	Yes	No	Yes
Set	No	Yes	No
Dictionary	Yes	Yes	Keys Unique

14. Real-Life Examples

Data Type	Real-Life Example
int	Age
float	Product price
string	Person name
bool	Login status
list	Shopping items
tuple	Weekdays
set	Unique followers
dict	Student database